Chapter 5

Rational Drug Use, Formulary Management, Pharmaceutical Care/ Medication Therapy Management/ Pharmacists' Patient Care Process

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Learning Objectives:

Objective 5.1 Define drug/medicine, medication, and health technology.
Objective 5.2 Overview rational drug use in clinical practice.
Objective 5.3 Detail the constitution and functions of a drug and therapeutics committee.
Objective 5.4 Discuss essential medicines policy and formulary management.
Objective 5.5 Explain methods in drug use evaluation (drug utilization review).
Objective 5.6 Explain methods to standardize pharmacy education and practice: pharmaceutical care, medication therapy management, CAPE educational outcomes, Nanjing Statement, pharmacists' patient care process, and entrustable professional activities.

OBJECTIVE 5.1. DEFINE DRUG/MEDICINE, MEDICATION, AND HEALTH TECHNOLOGY

A drug is defined with slight differences in each country's drug laws. For healthcare professionals, drug is synonymous with medicine. Especially as "medicine" is also used to represent the practice of physicians, use of "drug" is preferred for pharmacists in professional communications. For the public, the term "drug" is commonly used as a substance of abuse; use of the term "medicine" is preferred for pharmacists while communicating with the public. Many agencies, including the World Health Organization (WHO), recommend using the term medicine. In healthcare regulations of many countries, the term drug is still commonly used for medicinal products. In the United States, the drug regulatory agency is called the Food and Drug Administration (FDA), whereas in the European Union, it is called the European Medicines Agency.

The US FDA defines a drug as follows:

- A substance recognized by an official pharmacopeia or formulary.
- A substance intended for use in the diagnosis, cure, mitigation, treatment, or prevention of disease.
- A substance (other than food) intended to affect the structure or any function of the body.
- A substance intended for use as a component of a drug but not a device or a component, part or accessory of a device.
- Biological products are included within this definition and are covered by the same laws and regulations, but differences exist regarding their manufacturing processes (chemical process vs. biological process).[1]

A newer way of defining all pharmaceutical products together is by calling them "health technology." This includes drugs and medical devices. WHO defines a medical device as any instrument, apparatus, implement, machine, appliance,

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implant, reagent for in vitro use, software, material, or other similar or related article, intended by the manufacturer to be used, alone or in combination, for human beings, for one or more of the specific medical purpose(s) of:

- diagnosis, prevention, monitoring, treatment, or alleviation of disease,
- diagnosis, monitoring, treatment, alleviation of, or compensation for an injury,
- investigation, replacement, modification, or support of the anatomy or of a physiological process,
- supporting or sustaining life,
- control of conception,
- disinfection of medical devices,
- providing information by means of in vitro examination of specimens derived from the human body.

Moreover, device does not achieve its primary intended action by pharmacological, immunological, or metabolic means, in or on the human body, but may be assisted in its intended function by such means.

Note: Products that may be considered to be medical devices in some jurisdictions but not in others include the following:

- disinfection substances,
- aids for persons with disabilities,
- devices incorporating animal and human tissues, and
- devices for in vitro fertilization or assisted reproduction technologies.[2]

Drug Names

Drugs have generic names and brand names. For example, atorvastatin is the generic name and Lipitor is a brand name. Brand names are assigned by the pharmaceutical companies, and the generic name is selected by WHO. Pharmaceutical companies keep both brand and generic names on the product labels. The brand name is also called a proprietary name, and the generic name is an International Nonproprietary Name.[3] When a company develops and releases a new drug into the market, they will have market exclusivity protected by patent law. Once the patent expires, generic or branded versions of the same bioequivalent product could be released by any companies following approval by an appropriate regulatory agency. If a drug is marketed by the generic name, without a brand name, it is called as a generic product. The generic version of a biological product is called a biosimilar.[4] When we write these names within a sentence, a brand name starts with a capital letter and generic names start with a small letter. Another unique name for the pharmacological product is International Union of Pure and Applied Chemistry (IUPAC) name, a detailed chemical name.[5] Aspirin was originally the brand name of acetylsalicylic acid, but it is now commonly used as its generic name. The generic name is the same for most drugs throughout the world. However, there are exceptions. In the United States, the generic name for paracetamol is acetaminophen, salbutamol is called albuterol, and pethidine is meperidine. The problem does not stop here. In different languages such as German, Spanish, etc., the names are spelled and pronounced differently. To solve this problem, WHO assign a unique international code for each drug as per the Anatomical Therapeutic and Chemical (ATC) classification system. Some drugs may have more than one ATC code for their different therapeutic uses. For ibuprofen, the ATC code is "M01AE01." "M" stands for the musculoskeletal anatomical system, "01A" stands for nonsteroidal antiinflammatory and antirheumatic therapeutic class, "E" stands for propionic acid derivative chemical class, and the last number "01" stands for ibuprofen differentiating it from other nonsteroidal antiinflammatory drugs (NSAIDs) in the class.[6]

OBJECTIVE 5.2. OVERVIEW RATIONAL DRUG USE IN CLINICAL PRACTICE

Rational use of drugs is defined by WHO as "patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them and their community."[7]

Some of the WHO/International Network for the Rational Use of Drug (INRUD) drug use indicators for primary healthcare facilities (WHO, 1993) are as follows:

Prescribing indicators:

- Average number of drugs prescribed per patient encounter
- % drugs prescribed by generic name
- % encounters with an antibiotic prescribed
- % encounters with an injection prescribed
- % drugs prescribed from an essential drugs list or formulary

Patient care indicators:

- Average consultation time
- Average dispensing time
- % drugs actually dispensed
- % drugs adequately labeled
- % patients with knowledge of correct doses

Facility indicators:

- Availability of essential drugs list or formulary to practitioners
- Availability of clinical guidelines
- % key drugs available

Complementary drug use indicators:

- Average drug cost per encounter
- % prescriptions in accordance with clinical guidelines.[7]

These indicators are usually assessed by drug use evaluations (DUEs). Irrational drug use may be any clinically significant inappropriate use of drugs such as no drugs when required, unnecessary drugs, overuse, underuse, ineffective drugs, unsafe drugs, etc. Irrational drug use can be identified, prevented, or managed by the methods developed through pharmacy practice. A drug and therapeutics committee (DTC) could function to improve the rational use of drugs in a hospital.

WHO advocated 12 key interventions to promote more rational use:

- Establishment of a multidisciplinary national body to coordinate policies on drug use
- Use of clinical guidelines
- Development and use of a national essential drugs list
- Establishment of drug and therapeutics committees in districts and hospitals
- Inclusion of problem-based pharmacotherapy training in undergraduate curricula
- Continuing in-service medical education as a licensure requirement
- Supervision, audit, and feedback
- Use of independent information on drugs
- Public education about medicines
- Avoidance of perverse financial incentives
- Use of appropriate and enforced regulation
- Sufficient government expenditure to ensure availability of drugs and staff.[7]

OBJECTIVE 5.3. DETAIL THE CONSTITUTION AND FUNCTIONS OF A DRUG AND THERAPEUTICS COMMITTEE

DTC, also known as pharmacy and therapeutics committee (PTC), functions typically in institutional (hospital) settings. Similar committees could also work at public health (governmental and nongovernmental agencies) levels to develop state/national formularies and advice on the rational management of drugs in the region. DTC formulates policies and guidelines for the rational use of drugs and other pharmaceutical products. WHO uses the term DTC, whereas in the United States PTC is more commonly used.

According to WHO, goals and objectives of the DTC are as follows:

The goal of a DTC is to ensure that patients are provided with the best possible cost-effective and quality of care through determining what drugs will be available, at what cost, and how they will be used.

To achieve this goal, a DTC will have the following objectives:

- to develop and implement an efficient and cost-effective formulary system that includes consistent standard treatment protocols, a formulary list, and formulary manual
- to ensure that only efficacious, safe, cost-effective, and best quality drugs are used

- to ensure the best possible drug safety through monitoring, evaluating, and thereby preventing, as far as possible, adverse drug reactions (ADRs) and medication errors
- to develop and implement interventions to improve drug use by prescribers, dispensers, and patients; this will require the investigation and monitoring of drug use.[8]

Constitution of Drug and Therapeutics Committee

The DTC is usually constituted with expected competency to perform its duties. Any committee could have odd numbers of members with voting powers that help in avoiding a draw. A DTC should include stakeholders from major clinical specialty departments, administrations, and pharmacies. A small committee could be formed with a minimum of seven members (2−3 doctors, 1−2 pharmacists, 1 nurse, 1 administrator, and another member as per the hospital's requirement, preferably a medical IT specialist). The number shall vary based on the hospital's organizational structure. The following points should be considered to decide the committee to be small or large. For a small hospital, a small committee might be suitable. Small committees are less expensive and more easily reach consensus decisions.

Large committees are suitable for large hospitals with extensive pharmacy operations:

- Greater expertise by adding professionals with different backgrounds.
- Less workload for each member.
- Involving more of the hospital staff helps with the implementation of decisions.

The following members could be included based on defined responsibilities:

- a representative clinician from each major specialty, including surgery, obstetrics and gynecology, internal medicine, pediatrics, infectious diseases, and general practice (to represent the community)
- a clinical pharmacist
- a nurse, usually the senior infection control nurse, or nursing superintendent
- a pharmacist (usually the chief or deputy chief pharmacist) or a pharmacy technician where there is no pharmacist
- an administrator, representing the hospital administration and finance department
- a clinical microbiologist or a laboratory technician where there is no microbiologist
- a member of the hospital records department.[8]

Some DTCs also include drug information pharmacists, trainees, consumer representatives, etc.

Common subcommittees of DTC are the infection control committee, procurement committee, patient safety committee, etc.

Functions of Drug and Therapeutics Committee

DTC teams up with other committees and professionals in the hospital to manage the purchase, inventory, distribution, rational use, and disposal of pharmaceutical products in the hospital. DTC's functions are mostly advisory. It also develops policies and guidelines and plans strategies to implement change. Direct patient care is the responsibility of clinical staff and the DTC only monitors and suggests how to improve it. Some of the key functions of DTC are as follows:

- Advice healthcare professionals, administrators, and other committees on any issues related to pharmaceutical products. DTC has a pool of experts with technical expertise that could prepare guidance documents combining evidence, local experience, patient values, and technological support available. DTC promotes evidence-based clinical practices.
- Develop drug policies and plan strategies to implement new policies. The policies addressed generally include purchase, inventory, distribution, rational use, and disposal of pharmaceutical products. Examples of some specific policies include reduce the drug budget, procure drugs only from manufacturers that follow good manufacturing practices and holding the WHO prequalification certificate, hospital formulary management, prescribe only from the hospital formulary and in generic names of drugs, pharmacovigilance practices, safety culture, medical reconciliation in transitions of care, return or dispose expired drugs without polluting groundwater or air, etc.
- Prepare essential drugs list/hospital formulary and standard treatment guidelines (STGs)/clinical protocols.
- Perform DUE studies of drug-related problems such as medication errors and ADRs and suggest improving better utilization of resources for rational drug use.
- Conduct pharmacoeconomics evaluations to assure cost-effective and affordable drug use in the hospital.

- Monitor information management system in the hospital to protect patient privacy and confidentiality in the system. The DTC could also monitor potential legal and ethical violations in clinical practice in collaboration with other bodies in the hospital.
- The DTC also works with the quality assurance committee to ensure the hospital follows international patient safety goals and accreditation standards. It also self-evaluates its performance to deliver better services to the hospital and the community.[8]

For a DTC to function, it should have a multidisciplinary, transparent approach, technical competence, and an official mandate. Being multidisciplinary reflects different professionals and departments that have the competency to advise other professionals or bodies in the hospital. A collaborative clinical practice work environment requires teamwork between different healthcare professionals. The transparent approach is essential for providing a clear understanding of the situation by unbiased assessment and decision making. Finally, the official mandate is essential to implement a change. Otherwise, advice shall be ignored by the workforce. (E.g., if the prescribers ignore the advice to prescribe in generic names, the committee should have the support of the hospital administrator to warn or impose the advice as a rule.) The committee is for this reason inclusive of administrative staff. This does not mean the committee is adamant on its decisions but is sensitive to recognize practice difficulties and suggest relaxations of or revisions to the advice.

OBJECTIVE 5.4. DISCUSS ESSENTIAL MEDICINES POLICY AND FORMULARY MANAGEMENT

In this section, to have consistency the term medicine is used under this objective instead of drug. Essential medicines policy and formulary management principles and procedures are promoted by WHO. WHO defines essential medicines as those that satisfy the priority healthcare needs of the population. The essential medicines list is also called a formulary list. They are selected with due regard to public health relevance, evidence on efficacy and safety, and comparative cost-effectiveness. Essential medicines are intended to be available within the context of functioning health systems at all times in adequate amounts, in the appropriate dosage forms, with assured quality and adequate information, and at a price the individual and the community can afford.

The implementation of the concept of essential medicines is intended to be flexible and adaptable to many different situations; exactly which medicines are regarded as essential remains a national responsibility. WHO essential medicines list is a model that countries use in making their essential list. Hospital formularies are made by hospitals referring to the state/country/WHO formularies. For a hospital, the more relevant list of medicines is the hospital formulary that is made to address the healthcare needs of the majority population that the hospital serves. Regional variations and nature of hospital policies (corporate, not-for-profit, etc.) shall result in significant variations in the population availing its services. A corporate and charity hospital in the same city shall have a significantly different essential medicines list or formulary.

It is difficult to achieve efficiency in the hospital pharmaceutical system if there are too many medicines. All aspects of medicines management, including procurement, storage, distribution, and use, are easier if fewer items are managed.

The WHO Model List of Essential Medicine (EML), which serves as a guide for the development of national and institutional essential medicine lists, is updated and revised every 2 years by the WHO Expert Committee on Selection and Use of Medicines. The EML was first published in 1977 and the Model List of Essential Medicines for Children (EMLc) was first published in 2007.[9]

The 20th EML and sixth EMLc, published on June 6, 2017, mark the 40th anniversary of this flagship WHO tool to expand access to medicines. The 20th EML adds 30 medicines for adults and 25 for children and specifies new uses for 9 already-listed products, bringing the total to 433 medicines deemed essential for addressing the most important public health needs globally.[10]

Most countries have national lists, and some have provincial or state lists as well. National lists of essential medicines guide the procurement and supply of medicines in the public sector, schemes that reimburse medicine costs, medicine donations, and local medicine production.

WHO Criteria for Selection of Essential Medicines

The WHO Expert Committee on Selection and Use of Essential Medicines meets every 2 years to review the latest scientific evidence on the efficacy, safety, and cost-effectiveness of medicines to revise and update the WHO EML and EMLc. Committee members are selected from WHO Expert Advisory Panels based on equitable geographical representation, gender balance, and professional competencies to provide a representation of different approaches and practical experience from all regions of the world.

Essential medicines are selected with due regard to disease prevalence and public health relevance, evidence of clinical efficacy and safety, and comparative costs and cost-effectiveness.

Which treatment is recommended and which medicines are selected depend on many factors, such as the pattern of prevalent diseases, treatment facilities, the training and experience of available personnel, financial resources, and genetic, demographic, and environmental factors. The following criteria are used by the WHO Expert Committee on the Selection and Use of Essential Medicines:

- Only medicines for which sound and adequate evidence of efficacy and safety in a variety of settings is available should be selected
- Relative cost-effectiveness is a major consideration for choosing medicines within the same therapeutic category. In comparisons between medicines, the total cost of the treatment—not only the unit cost of the medicine—must be considered and compared with its efficacy
- In some cases, the choice may also be influenced by other factors such as pharmacokinetic properties or by local considerations such as the availability of facilities for manufacture or storage
- Each medicine selected must be available in a form in which adequate quality, including bioavailability, can be ensured; its stability under the anticipated conditions of storage and use must be determined
- Most essential medicines should be formulated as single compounds. Fixed dose combination products are selected only when the combination has a proven advantage in therapeutic effect, safety, adherence, or in decreasing the emergence of drug resistance in malaria, tuberculosis (TB), and HIV/AIDS.[10]

Formulary

As per definition, a *formulary list* and essential medicines list convey the same meaning. They are the medicines that satisfy the priority healthcare needs of the population. The *formulary manual* (e.g., WHO Model Formulary, British National Formulary, hospital formulary) is usually published as a small book that has the essential medicines list with short clinical profiles of each medicine. It serves as a medicines information resource for the healthcare professionals in their clinical practice.

The formulary process or formulary system is the cornerstone of good pharmaceutical management and rational medicine use. It consists of preparing, using, and updating a formulary list (essential medicines list, EML also known as essential drug list, EDL), a formulary manual (providing information on medicines in the formulary list), and STGs. Choosing the most appropriate therapies and selecting the most cost-effective good-quality medicines lead to better quality of care and more efficient, equitable use of resources.[8]

Since its first publication in 2002, the WHO Model Formulary has become a source of independent information on essential medicines for pharmaceutical policy-makers and prescribers worldwide. For each medicine, the formulary provides information on use, dosage, adverse effects, contraindications, and warnings, supplemented by guidance on selecting the right medicine for a range of conditions.[11]

OBJECTIVE 5.5. EXPLAIN METHODS IN DRUG USE EVALUATION (DRUG UTILIZATION REVIEW)

DUE (also known as drug utilization review or drug utilization research) tries to interpret the quality and quantity indicators to understand and improve the use of pharmaceuticals at a clinical level to international levels in healthcare system management. DUE is one of the fundamental research activities in pharmacy practice, and most of the research performed in pharmacy practice refers to DUE. Advanced research fields such as pharmacoepidemiology, pharmacovigilance, pharmacoeconomics, outcomes research, etc., claim to produce more useful knowledge compared with DUE. These newer fields of research evolved from DUE, and they use some DUE methods at least in part. The DUE studies by itself are still commonly used as healthcare system indicators locally and internationally.

Practice-based data can be used to check the quality of prescribing, dispensing, administering, etc., and quantities of pharmaceutical use in a healthcare system (hospital, community, country, etc.). Prescribing patterns are commonly used methods to check adherence of prescribers to standard treatment guidelines and the rational or good prescribing practices in general. Qualities of drug dispensing procedures and pharmacist interventions that prevent medication errors at the prescriber, nurse, patient, or pharmacist level can be assessed by DUE research. Hospital administrators or governmental health authorities might need to know the usage patterns of drugs that are subsidized or reimbursed. Economic aspects, including the drug budget management, are largely conducted as pharmacoeconomic studies. DUEs are commonly used in resource-limited settings in budget planning and assessments. In hospitals, the DTC typically organizes DUEs for the formulary management.

Defined daily dose (DDD) is the most regarded drug utilization metric. It was developed as a universal measure of drug consumption. Expression of consumption regarding cost or numbers is not comparable between countries. DU90% is another metric used to identify the most commonly used drugs. It measures the number of drugs accounting for 90% of the use in DDDs. An antimicrobial day (also known as Day Of Therapy-DOT) is defined by any amount of a specific antimicrobial agent administered in a calendar day to a particular hospitalized patient. DOT is a useful measure in antimicrobial stewardship.

DDD is defined as *the assumed average maintenance dose per day for a drug used for its main indication in adults*. DDDs are assigned to drugs that have an ATC code.[12] DDDs are assigned for drugs that are approved in at least one country. DDDs are not assigned for radiopharmaceuticals, appetite stimulants, anesthetics, blood substitutes, parenteral organic nitrates, many topical products, vaccines, antineoplastic agents, etc., due to high dose variation. The DDD is a unit of measurement and does not necessarily correspond to the recommended or prescribed daily dose. Drug utilization data expressed using DDDs are a rough estimate of drug use that could vary in different patient populations with comorbidities.[13]

DDDs are assigned by the WHO Collaborating Centre for Drug Statistics Methodology in the Norwegian Institute of Public Health and funded by the Norwegian government. An assigned DDD may be revised after 3 years. Drug utilization figures should ideally be presented using a relevant denominator in the health context, such as numbers of DDDs per 1000 inhabitants per day, DDDs per inhabitant per year, or as DDDs per 100 bed days.

For a patient, the actual dose used is the *DDD*. If numbers of days in that treatment period also need to be noted, the *DDD/patient* could be used. Once the WHO-assigned DDD is available for a drug, use of that drug by an individual regarding DDD could be calculated as follows:

$$\text{Drug usage (DDD) for a patient} = \frac{(\text{Item issues} \times \text{amount of drug per item})}{\text{DDD}}$$

DDDs per 1000 inhabitants per day: Sales or prescription data presented in DDDs per 1000 inhabitants per day may provide a rough estimate of the proportion of the study population treated daily with a particular drug or group of drugs.

DDDs per 100 bed days: The DDDs per 100 bed days may be applied when drug use by inpatients is considered. The common definition of a bed day is "*a day during which a person is confined to a bed and in which the patient stays overnight in a hospital.*"[14]

OBJECTIVE 5.6. EXPLAIN METHODS TO STANDARDIZE PHARMACY EDUCATION AND PRACTICE: PHARMACEUTICAL CARE, MEDICATION THERAPY MANAGEMENT, CAPE EDUCATIONAL OUTCOMES, NANJING STATEMENT, PHARMACISTS' PATIENT CARE PROCESS, AND ENTRUSTABLE PROFESSIONAL ACTIVITIES

Pharmaceutical Care

The pharmaceutical care concept revolutionized the practice of pharmacists toward delivering more services to patients in addition to the traditional dispensing of medication. The concept emerged in the mid-1970s. Globally, pharmaceutical care services are provided by pharmacists in different forms.

Pharmaceutical care is the leading term used globally to represent overall patient care services, including the WHO and the International Pharmaceutical Federation (FIP). In Australia, the related term is Quality Use of Medicines (QUM). QUM is one of the central objectives of Australia's National Medicines Policy. QUM means selecting management options wisely, choosing a suitable medicine if a medicine is considered necessary, and using medicines safely and effectively. QUM applies to medicines used at both patient and population levels.[15]

In the United States the term used mostly is medication therapy management (MTM), but pharmaceutical care is also used. One of the biggest problems of pharmaceutical care or MTM is that it is not standardized; many define and use it in different ways. In 2014, in the United States, a collaborative effort of leading pharmacy organizations resulted in a standardized method to deliver care services called the pharmacists' patient care process (PPCP). PPCP combined principles of pharmaceutical care and MTM.[16]

Definitions of Pharmaceutical Care

The original and most popular definition of pharmaceutical care was developed by Charles D. Hepler and Linda M. Strand in 1990. The Hepler and Strand definition is:

> *Pharmaceutical care is the responsible provision of drug therapy for the purpose of achieving definite outcomes that improve a patient's quality of life.*

These outcomes are as follows:

- curing a disease,
- eliminating or reducting a patient's symptomatology,
- arresting or slowing down a disease process, or
- preventing a disease or symptomatology.

Three major functions of pharmaceutical care are to

- identify potential and actual drug-related problems
- resolve actual drug-related problems, and
- prevent drug-related problems.[17]

In 1992, the American Society for Health-System (ASHP) Pharmacists adopted the Hepler and Strand definition with two changes. The ASHP definition is:

> *Pharmaceutical care is the direct, responsible provision of medication-related care for the purpose of achieving definite outcomes that improve a patient's quality of life.*[18]

In 1998, the FIP adopted the Hepler and Strand definition with a little modification:

> *Pharmaceutical care is the responsible provision of drug therapy for the purpose of achieving definite outcomes that improve or maintain a patient's quality of life.*[19]

The addition to *maintain* a patient's quality of life is significant when improvement is not possible. Pharmacists, in collaboration with other healthcare professionals, aim to at least maintain individuals' quality of life. Suffering could be decreased even if a cure is not possible.

Another significant effort came from the Pharmaceutical Care Network Europe (PCNE) in 2013. The PCNE definition is:

> *Pharmaceutical Care is the pharmacist's contribution to the care of individuals in order to optimize medicines use and improve health outcomes.*[20]

Pharmaceutical care practice consists of three major components: a philosophy of practice, a patient care process, and a practice management system.

The philosophy of pharmaceutical care practice consists of

- a description of the social need for the practice,
- a clear statement of individual practitioner responsibilities to meet this social need,
- the expectation to be patient centered, and
- the requirement to function within the caring paradigm.

These four constructs, social need, practitioner responsibilities, patient-centered approach, and caring paradigm, formulate professionalism for pharmacists.[21]

Steps in pharmaceutical care vary widely. Many develop their suitable models of the care process. What is common in the process is to collect relevant information, assess the information for healthcare needs and drug therapy problems, prepare a care plan, and evaluate the outcomes. Alternatively, it can be identify, assess, resolve, and monitor drug-related problems. Repeat the process if required. Some of the common formats to report a pharmaceutical care intervention are SOAP (subjective, objective, assessment, plan), TITIRS (title, introduction, text, recommendation, signature), and FARM (findings, assessment, recommendations/resolutions, management). Among these, SOAP notes are the most common.[22]

Hepler, who originally defined pharmaceutical care, suggested eight steps in the pharmaceutical care process:

1. Record and interpret relevant patient information.
   What do we need to know about this patient?
2. Document Therapeutic Plan and Desired Therapeutic Objectives for the Patient.
   What do we intend to achieve with this therapy in this patient?
3. Evaluate a Therapeutic Plan.
   Is this an acceptable plan to achieve those objectives for this patient?
4. Design a Monitoring Plan.
   What evidence will we need to assess the progress of therapy?

5. Dispense, Drug Products, Advise Patient.
   Can this patient (or family caregiver, etc.) now make the best use of this drug?
6. Implement the Monitoring Plan.
   What evidence do I need to assist in evaluating this patient's progress?
7. Identify Possible Drug Therapy Problems.
   Is this patient progressing toward therapeutic objectives? Are there indicators of drug therapy problems?
8. Respond to Problems.
   What action should I take now?[23]

ASHP categorized drug therapy problems into 11 types for the national skills competition in 2012:

1. Correlation between drug therapy and medical problems
2. Appropriate drug selection
3. Drug regimen
4. Therapeutic duplication
5. Drug allergy or intolerance
6. Adverse drug events
7. Interactions: drug−drug, drug−disease, drug−nutrient, and drug−laboratory test
8. Social or recreational drug use
9. Failure to receive therapy
10. Financial impact
11. Patient knowledge of drug therapy

As per PCNE, a drug-related problem is an event or circumstance involving drug therapy that actually or potentially interferes with desired health outcomes. PCNE drug-related problems are classified in Table 5.1.

Medication Therapy Management

Following the late 2003 passage of the Medicare Prescription Drug Improvement and Modernization Act, a unanimous definition of MTM was required in the United States. Eleven pharmacy organizations in the US coordinated by the American Pharmacists Association (APhA) approved a definition in 2004.[25] The MTM definition is:

> *MTM is a service or group of services that optimize therapeutic outcomes for individual patients. MTM services include medication therapy reviews, pharmacotherapy consults, anticoagulation management, immunizations, health and wellness programs and many other clinical services. Pharmacists provide MTM to help patients get the best benefits from their medications by actively managing drug therapy and by identifying, preventing and resolving medication-related problems.*[26]

MTM is a distinct service or group of services that optimizes therapeutic outcomes for individual patients.

MTM services are independent of, but can occur in conjunction with, the provision of a medical product.

MTM encompasses a broad range of professional activities and responsibilities within the licensed pharmacist's, or other qualified healthcare provider's, scope of practice. These services include, but are not limited to, the following, according to the individual needs of the patient:

- Performing or obtaining necessary assessments of the patient's health status
- Formulating a medication treatment plan
- Selecting, initiating, modifying, or administering medication therapy
- Monitoring and evaluating the patient's response to therapy, including safety and effectiveness
- Performing a comprehensive medication review to identify, resolve, and prevent medication-related problems, including adverse drug events
- Documenting the care delivered and communicating essential information to the patient's other primary care providers
- Providing verbal education and training designed to enhance patient understanding and appropriate use of his/her medications
- Providing information, support services, and resources designed to enhance patient adherence to his/her therapeutic regimens
- Coordinating and integrating medication therapy management services within the broader healthcare management services being provided to the patient.[27]

TABLE 5.1 Pharmaceutical Care Network Europe Classified Drug-Related Problems (DRPs), Causes, Interventions, and Outcomes V8.0[24]; the Basic Classification

	Code V8.0	Primary Domains
Problems (also potential)	**P1**	**Treatment effectiveness** There is a (potential) problem with the (lack of) effect of the pharmacotherapy
	P2	**Treatment safety** Patient suffers, or could suffer, from an adverse drug event
	P3	**Others**
Causes (including possible causes for potential problems)	**C1**	**Drug selection** The cause of the DRP can be related to the selection of the drug
	C2	**Drug form** The cause of the DRP is related to the selection of the drug form
	C3	**Dose selection** The cause of the DRP can be related to the selection of the dosage schedule
	C4	**Treatment duration** The cause of the DRP is related to the duration of treatment
	C5	**Dispensing** The cause of the DRP can be related to the logistics of the prescribing and dispensing process
	C6	**Drug use/process** The cause of the DRP is related to the way the patient gets the drug administered by a health professional or carer, in spite of proper instructions (on the label)
	C7	**Patient related** The cause of the DRP can be related to the patient and his/her behavior (intentional or nonintentional)
	C8	**Other**
Planned interventions	**I0**	**No intervention**
	I1	**At prescriber level**
	I2	**At patient level**
	I3	**At drug level**
	I4	**Other**
Intervention acceptance	**A1**	**Intervention accepted**
	A2	**Intervention not accepted**
	A3	**Other**
Status of the DRP	**O0**	**Problem status unknown**
	O1	**Problem solved**
	O2	**Problem partially solved**
	O3	**Problem not solved**

As per APhA, a program that provides coverage for MTM services shall include:

1. Patient-specific and individualized services or sets of services provided directly by a pharmacist to the patient. These services are distinct from formulary development and use, generalized patient education and information activities, and other population-focused quality-assurance measures for medication use.

2. Face-to-face interaction between the patient and the pharmacist as the preferred method of delivery. When patient-specific barriers to face-to-face communication exist, patients shall have equal access to appropriate alternative delivery methods. MTM programs shall include structures supporting the establishment and maintenance of the patient/pharmacist relationship.
3. Opportunities for pharmacists and other qualified healthcare providers to identify patients who should receive MTM services.
4. Payment for MTM services consistent with contemporary provider payment rates that are based on the time, clinical intensity, and resources required to provide services.
5. Processes to improve continuity of care, outcomes, and outcome measures.[28]

Core Elements of an MTM Service Model in Pharmacy Practice

The MTM service model in pharmacy practice includes the following five core elements:

- Medication therapy review
- Personal medication record
- Medication-related action plan
- Intervention and/or referral
- Documentation and follow-up[28]

A variety of direct patient care services offered by pharmacists through MTM services include comprehensive medication reviews (CMRs) and targeted medication reviews (TMRs). CMRs consist of a full review of all medications and past medication history. Following a CMR and when a medication-related problem is identified, a TMR can be provided to resolve the specific finding. CMRs and TMRs are services delivered by pharmacists to optimize patient-specific medication regimens to improve patient outcomes and reduce treatment-related harm.

MTM services are provided to patients in a variety of clinical practice settings (i.e., inpatient, ambulatory care, managed care, long-term care, and telehealth). However, MTM services are frequently not billed or reimbursed and are related to numerous factors, including practice model limitations, lack of integrated billing support services, or payer-based confusion regarding benefit coverage. The shift from fee-for-service payment models to quality outcome-driven models as well struggles to include pharmacist providers within the standard scope of providers at the federal level are key factors that limit the full utilization of MTM services. Functional payment models for MTM services are essential to ensure the ability to achieve improvements in patient-specific and population-based health outcomes. Commercial insurers and government payers are not universally reimbursed for MTM services despite the fact that these services demonstrate a strong return-on-investment based on improving quality of health outcomes, improved treatment adherence, reduced adverse side effects, and other medication-related problems. Some payers, such as Kaiser Permanente, are implementing "at-risk" contracts in bundled practice models like Accountable Care Organizations to ensure improved provider engagement and economic performance.

Despite the overwhelming success of numerous demonstration projects, research studies, and practice model initiatives, including the Asheville Project, Ten Cities Challenge, Project Impact, and multiple others, viable reimbursement mechanisms remain one of the key barriers to nationwide MTM implementation and sustainability.

Rosenthal et al. describe MTM services in three definable levels of intensity. Level 1 (low intensity) involves adherence management, Level 2 (mid-level) involves drug-related problems, and Level 3 (high intensity) involves disease state management. Although Level 3 may be described as the ideal performance level, MTM services should be tailored to meet the needs of each specific patient, family, or caregiver. Table 5.2 describes the goals of these three levels of performance.[29,30]

Just as in medical practice, recognizing differences in pharmacist practitioners' level of training, clinical proficiencies, practice experience, and credentialing may also help in driving optimal levels of performance for targeted patient populations.[31]

A broader array of clinical service coding billing methods will assist in improving future reimbursement rates for pharmacists' services.[32] There will always be opportunities for future innovation and practice enhancement at both the state and federal levels to improve the utilization of pharmacists' patient care services. Pharmacists remain diligent and committed to expanding collaboration with other healthcare providers to ensure optimization of medication therapy to improve the quality of health outcomes, patient safety, and cost-effectiveness of future healthcare delivery models.[33]

TABLE 5.2 Performance Levels of Medication Therapy Management

Performance Level	Goals
Level 1	Adherence management
Level 2	Interventions in drug-related problems
Level 3	Disease state management

Nanjing Statements

The FIP's Nanjing Statements were released in 2017. The Nanjing Statements are intended for education providers, including Schools of Pharmacy and providers of Continuing Professional Development and Continuing Education. They are to be used for self-assessment and monitoring (at the country level or the education provider level), identification of gaps and strategic planning, and improving the process of education. The statements have eight clusters as follows:[34]

- Shared Global Vision: A shared global vision promotes workforce development in the context of pharmaceutical education and training. This global vision should help professional leadership bodies, educators, and regulators in developing a national or regional vision based on the priorities and resources of the country or region, with the aim of developing new drugs and improving their use for better health.
- Professional Skills Mix: Pharmacists in all settings and pharmaceutical scientists need competence, skills, knowledge, and attitudes to meet the needs of the public and interact with other healthcare professionals.
- Recruitment of Students: Recruiting students who have a profile that fits the requirements of the school and are aligned with the profile of pharmacists desired for the country.
- Foundation Training and Leadership: Foundation training includes the process of education and leadership development for students and new graduates in pharmacy and the pharmaceutical sciences with a priority on developing the next generation of clinical, scientific, academic, and professional leaders.
- Experiential Education: Experiential education programs are where students incrementally develop their pharmacy practice and science skills in a wide variety of real-life settings.
- Resources and Academic Staff: Resources and academic staff refer to equipment, finances, technology, and human resources needed to prepare pharmacists and pharmaceutical scientists properly.
- Quality Assurance: Quality assurance refers to the key aspects and mechanisms to identify opportunities for and make improvement in pharmacy and pharmaceutical sciences education to ensure a good, sustainable performance and suitable competencies of the future workforce.
- Continuing Professional Development: Continuing professional development (CPD) refers to building on previous education as a pharmacist and pharmaceutical scientist.[34]

CAPE Educational Outcomes

Center for the Advancement of Pharmacy Education (CAPE) Educational Outcomes is a model used to plan pharmacy curricula. The CAPE Educational Outcomes could be used as a basis to develop learning objectives of the pharmacy undergraduate program.[35] The CAPE Educational Outcomes 2013 consists of the following:

Domain 1—Foundational Knowledge
1.1. Learner (Learner)
Domain 2—Essentials for Practice and Care
2.1. Patient-centered care (Caregiver)
2.2. Medication use systems management (Manager)
2.3. Health and wellness (Promoter)
2.4. Population-based care (Provider)

Domain 3—Approach to Practice and Care
3.1. Problem-solving (Problem solver)
3.2. Educator (Educator)
3.3. Patient advocacy (Advocate)
3.4. Interprofessional collaboration (Collaborator)
3.5. Cultural sensitivity (Includer)
3.6. Communication (Communicator)
Domain 4—Personal and Professional Development
4.1. Self-awareness (Self-aware)
4.2. Leadership (Leader)
4.3. Innovation and entrepreneurship (Innovator)
4.4. Professionalism (Professional)

Pharmacists' Patient Care Process

PPCP is a newer care process that combines pharmaceutical care and MTM. The Joint Commission of Pharmacy Practitioners (JCPP) is a forum of leading pharmacy organizations in the United States. PPCP was a group effort of these agencies in producing a standardized patient care process by pharmacists for pharmacists. It is a comprehensive approach to patient-centered care delivered by pharmacists in collaboration with other healthcare professionals. PPCP is based on the existing evidence of pharmaceutical care and MTM. It tries to incorporate the following concepts into patient care provided by the pharmacists:

- Quality care: Assuring quality in the healthcare products and services. Providing drugs of good quality and caring to prevent and manage any drug-related problems. Pharmacists regularly deliver evidence-based recommendations to healthcare professionals and patients.
- Access to healthcare: Being one of the most accessible healthcare professionals, pharmacists improve the patients' access to healthcare interventions.
- Cost-effective outcomes: Assuring value for money spent in achieving measurable clinical outcomes.
- Collaborative practice: Collaborating with physicians and other pharmacists to provide coordinated care.
- Patient-centered care: Giving importance to the role of patients in the clinical decision-making process (Fig. 5.1).

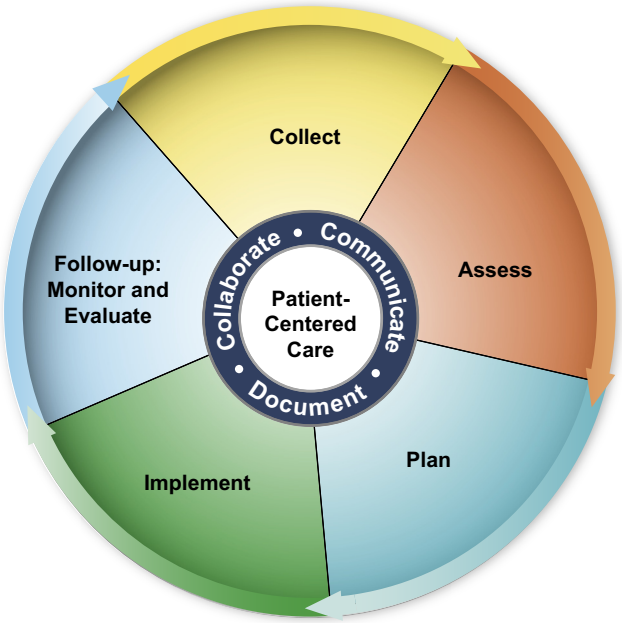


FIGURE 5.1 Pharmacists' patient care process (Joint Commission of Pharmacy Practitioners).[16]

As per the JCPP guidance, using principles of evidence-based practice, pharmacists perform the following:

Collect

The pharmacist assures the collection of necessary subjective and objective information about the patient to understand the relevant medical/medication history and clinical status of the patient. Information may be gathered and verified from multiple sources, including existing patient records, the patient, and other healthcare professionals. This process includes collecting

- A current medication list and medication use history for prescription and nonprescription medications, herbal products, and other dietary supplements.
- Relevant health data may include medical history, health and wellness information, biometric test results, and physical assessment findings.
- Patient lifestyle habits, preferences and beliefs, health and functional goals, and socioeconomic factors that affect access to medications and other aspects of care.

Assess

The pharmacist assesses the information collected and analyzes the clinical effects of the patient's therapy in the context of the patient's overall health goals to identify and prioritize problems and achieve optimal care. This process includes assessing

- Each medication for appropriateness, effectiveness, safety, and patient adherence.
- Health and functional status, risk factors, health data, cultural factors, health literacy, and access to medications or other aspects of care.
- Immunization status and the need for preventive care and other healthcare services, where appropriate.

Plan

The pharmacist develops an individualized patient-centered care plan in collaboration with other healthcare professionals and the patient or caregiver who is evidence based and cost-effective. This process includes establishing a care plan that

- Addresses medication-related problems and optimizes medication therapy.
- Sets goals of therapy for achieving clinical outcomes in the context of the patient's overall healthcare goals and access to care.
- Engages the patient through education, empowerment, and self-management.
- Supports care continuity, including follow-up and transitions of care as appropriate.

Implement

The pharmacist implements the care plan in collaboration with other healthcare professionals and the patient or caregiver. During the process of implementing the care plan, the pharmacist

- Addresses medication- and health-related problems and engages in preventive care strategies, including vaccine administration.
- Initiates, modifies, discontinues, or administers medication therapy as authorized.
- Provides education and self-management training to the patient or caregiver.
- Contributes to coordination of care, including the referral or transition of the patient to another healthcare professional.
- Schedules follow-up care as needed to achieve goals of therapy.

Follow-Up: Monitor and Evaluate

The pharmacist monitors and evaluates the effectiveness of the care plan and modifies the plan in collaboration with other healthcare professionals and the patient or caregiver as needed. This process includes the continuous monitoring and evaluation of

- Medication appropriateness, effectiveness, and safety and patient adherence through available health data, biometric test results, and patient feedback.

- Clinical endpoints that contribute to the patient's overall health.
- Outcomes of care, including progress toward or the achievement of goals of therapy.[16]

Core Entrustable Professional Activities

Core Entrustable Professional Activities (EPAs) for New Pharmacy Graduates are discrete, essential activities and tasks that all new pharmacy graduates must be able to perform without direct supervision on entering practice or postgraduate training. These statements were labeled as "core" to denote that these EPAs are expected of all graduates independent of practice setting. They serve as a baseline, not a ceiling.[36] The EPA Statements for New Pharmacy Graduates are as follows (Table 5.3):

TABLE 5.3 Core Entrustable Professional Activities for New Pharmacy Graduates[36]

Patient Care Provider Domain	Example Supporting Tasks
Collect information to identify a patient's medication-related problems and health-related needs.	• Collect a medical history from a patient or caregiver. • Collect a medication history from a patient or caregiver. • Discuss a patient's experience with medication. • Determine a patient's medication adherence. • Use health records to determine a patient's health-related needs relevant to the setting of care and the purpose of the encounter.
Analyze information to determine the effects of medication therapy, identify medication-related problems, and prioritize health-related needs.	• Assess a patient's signs and symptoms to determine whether the patient can be treated within the scope of practice or requires a referral. • Measure an adult patient's vital signs and interpret the results (e.g., body temperature, pulse rate, respiration rate, and blood pressure). • Interpret laboratory test results. • Identify drug interactions. • Perform a comprehensive medication review (CMR) for a patient. • Assess a patient's health literacy using a validated screening tool. • Compile a prioritized health-related problem list for a patient. • Evaluate an existing drug therapy regimen.
Establish patient-centered goals and create a care plan for a patient in collaboration with the patient, caregiver(s), and other health professionals that are evidence based and cost-effective.	• Follow an evidence-based disease management protocol. • Develop a treatment plan with a patient. • Manage drug interactions. • Select monitoring parameters to determine the therapeutic and adverse effects related to the treatment plan. • Determine the appropriate time interval(s) to collect monitoring data. • Create a patient-specific education plan.
Implement a care plan in collaboration with the patient, caregivers, and other health professionals.	• Write a note that documents the findings, recommendations, and plan from a patient encounter. • Educate a patient regarding the appropriate use of a new medication, device to administer a medication, or self-monitoring test. • Educate a patient on the use of medication adherence aids. • Assist a patient with behavior change (e.g., use shared decision-making and motivational strategies).
Follow-up and monitor a care plan.	• Collect monitoring data at the appropriate time interval(s). • Evaluate the selected monitoring parameters to determine the therapeutic and adverse effects related to the treatment plan.

Continued

TABLE 5.3 Core Entrustable Professional Activities for New Pharmacy Graduates[36]—cont'd

	• Recommend modifications or adjustments to an existing medication therapy regimen based on patient response. • Present a patient case to a colleague during a handoff or transition of care.
Interprofessional Team Member Domain:	**Example Supporting Tasks:**
Collaborate as a member of an interprofessional team.	• Contribute medication-related expertise to the team's work. • Explain to a patient, caregiver, or professional colleague each team member's role and responsibilities. • Communicate a patient's medication-related problem(s) to another health professional. • Use setting-appropriate communication skills when interacting with others. • Use consensus-building strategies to develop a shared plan of action.
Population Health Promoter Domain:	**Example Supporting Tasks:**
Identify patients at risk for prevalent diseases in a population.	• Perform a screening assessment to identify patients at risk for prevalent diseases in a population (e.g., hypertension, diabetes, depression).
Minimize adverse drug events and medication errors.	• Assist in the identification of underlying system-associated causes of errors. • Report adverse drug events and medication errors to stakeholders.
Maximize the appropriate use of medications in a population.	• Perform a medication use evaluation. • Apply cost–benefit, formulary, and/or epidemiology principles to medication-related decisions.
Ensure that patients have been immunized against vaccine-preventable diseases.	• Determine whether a patient is eligible for and has received recommended immunizations. • Administer and document recommended immunizations to an adult patient. • Perform basic life support.
Information Master Domain:	**Example Supporting Tasks:**
Educate patients and professional colleagues regarding the appropriate use of medications.	• Lead a discussion regarding a recently published research manuscript and its application to patient care. • Develop and deliver a brief (less than 1 h) educational program regarding medication therapy to health professional(s) or lay audiences.
Use evidence-based information to advance patient care.	• Retrieve and analyze scientific literature to make a patient-specific recommendation. • Retrieve and analyze scientific literature to answer a drug information question.
Practice Manager Domain:	**Example Supporting Tasks:**
Oversee the pharmacy operations for an assigned work shift.	• Implement pharmacy policies and procedures. • Supervise and coordinate the activities of pharmacy technicians and other support staff. • Assist in training pharmacy technicians and other support staff. • Assist in the evaluation of pharmacy technicians and other support staff. • Identify pharmacy service problems and/or medication safety issues. • Maintain the pharmacy inventory. • Assist in the management of a pharmacy budget. • Interpret pharmacy quality and productivity indicators using continuous improvement quality techniques. • Assist in the preparation for regulatory visits and inspections.

Continued

TABLE 5.3 Core Entrustable Professional Activities for New Pharmacy Graduates[36]—cont'd

Fulfill a medication order.	• Enter patient-specific information into an electronic health or pharmacy record system. • Prepare commonly prescribed medications that require basic sterile compounding or basic nonsterile compounding prior to patient use. • Determine if a medication is contraindicated for a patient. • Identify and manage drug interactions. • Determine the patient co-pay or price for a prescription. Ensure that formulary preferred medications are used when clinically appropriate. • Obtain authorization for a nonpreferred medication when clinically appropriate. • Assist a patient to acquire medication(s) through support programs.
Self-Developer Domain:	**Example Supporting Tasks:**
Create a written plan for continuous professional development.	• Create and update a curriculum vitae, resume, and/or professional portfolio. • Perform a self-evaluation to identify professional strengths and weaknesses.

Adapted with permission from the American Association of Colleges of Pharmacy. The full report is available at http://www.aacp.org/resources/education/cape/Pages/EPAs.aspx.

CONCLUSION

Services provided by the pharmacists are heavily based on rational drug use and other WHO guidance. Standards for improving practice and education of pharmacists are in place. Good examples of practice standards are PCNE classification of drug-related problems in Europe and PPCP in the United States